

GIT-HOL 4

Merge Conflict Resolution

Objective

- Resolve merge conflicts during branch merging.

Prerequisites

- Completed HOL 3.

Procedure

1. Create GitWork branch.
2. Create hello.xml and commit.
3. Switch to main and create hello.xml with different content.
4. Commit the changes.
5. Merge GitWork into main to generate conflict.
6. Resolve the conflict manually.
7. Commit the resolved file.
8. Update .gitignore and delete GitWork.

Commands

```
git checkout -b GitWork
echo "<message>Hello from GitWork</message>" > hello.xml
git add hello.xml
git commit -m "Added hello.xml in GitWork"
git checkout main
echo "<message>Hello from Main Branch</message>" > hello.xml
git add hello.xml
git commit -m "Added hello.xml in main"
git merge GitWork
git add hello.xml
git commit -m "Resolved merge conflict"
git branch -d GitWork
```

Output

```
MINGW64~/Users/lahar/GitDemo
lahar@laharitata MINGW64 ~/GitDemo (main)
$ git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git checkout -b GitWork
Switched to a new branch 'GitWork'

lahar@laharitata MINGW64 ~/GitDemo (GitWork)
$ echo "<message>Hello from GitWork</message>" > hello.xml

lahar@laharitata MINGW64 ~/GitDemo (GitWork)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it

lahar@laharitata MINGW64 ~/GitDemo (GitWork)
$ git commit -m "Added hello.xml in GitWork"
[GitWork 598d261] Added hello.xml in GitWork
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

lahar@laharitata MINGW64 ~/GitDemo (GitWork)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

lahar@laharitata MINGW64 ~/GitDemo (main)
$ echo "<message>Hello from Main Branch</message>" >
bash: syntax error near unexpected token 'newline'

lahar@laharitata MINGW64 ~/GitDemo (main)
$ echo "<message>Hello from Main Branch</message>" > hello.xml

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git add hello.xml
git commit -m "Added hello.xml in main"
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it
[main 44ba0fd] Added hello.xml in main
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git log --oneline --graph --decorate --all
* 44ba0fd (HEAD -> main) Added hello.xml in main
* 598d261 (GitWork) Added hello.xml in GitWork
//
* 84b5bea (origin/main) Added branch.txt in GitNewBranch
* 67dfc49 Added gitignore file
* 5e3d4eb Added welcome file

lahar@laharitata MINGW64 ~/GitDemo (main)
```

```
MINGW64~/Users/lahar/GitDemo
* 67dfc49 Added gitignore file
* 5e3d4eb Added welcome file

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git diff main GitWork
diff --git a/hello.xml b/hello.xml
index 064023b..84f3217 100644
--- a/hello.xml
+++ b/hello.xml
@@ -1,1 @@
-<message>Hello from Main Branch</message>
+<message>Hello from GitWork</message>

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

lahar@laharitata MINGW64 ~/GitDemo (main|MERGING)
$ git add hello.xml
git commit -m "Resolved merge conflict"
[main 9692828] Resolved merge conflict

lahar@laharitata MINGW64 ~/GitDemo (main)
$ notepad .gitignore

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git add .gitignore
git commit -m "Updated .gitignore"
[main 1934dc6] Updated .gitignore
1 file changed, 2 insertions(+), 1 deletion(-)

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git branch
  GitWork
* main

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git branch -d GitWork
Deleted branch GitWork (was 598d261).

lahar@laharitata MINGW64 ~/GitDemo (main)
$ git log --oneline --graph --decorate
* 1934dc6 (HEAD -> main) Updated .gitignore
* 9692828 Resolved merge conflict
//
| * 598d261 Added hello.xml in GitWork
| * 44ba0fd Added hello.xml in main
//
* 84b5bea (origin/main) Added branch.txt in GitNewBranch
* 67dfc49 Added gitignore file
* 5e3d4eb Added welcome file
```

Result

Merge conflict was resolved successfully and changes were committed.